THE LENGTH OF A QUASI-COHERENT MODULE AT A POINT IS DETECTED ON ANY ETALE CHART

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a strict henselization of the chart local ring, a base change formula for the stalk and flat base change of lengths

Editorial note. The label lemma-length names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The stalk of a quasi-coherent module at a geometric point of an algebraic space, expressed as a base change of the stalk on an etale chart
- The identification of the local ring at a geometric point with the strict henselization of the local ring of a chart, and the flatness of the chart map on local rings
- Flat base change of lengths of modules over local rings, and the identification of the quotient by the maximal ideal with the residue field

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Prerequisite pointer invoked at proof line(s) 501; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07QM: sources/stacks-chow/tags.txt, source lines 9837–9837. Prerequisite pointer invoked at proof line(s) 507; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04KF: sources/stacks-chow/tags.txt, source lines 5971–5971. Prerequisite pointer invoked at proof line(s) 498; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05VP: sources/stacks-chow/tags.txt, source lines 7558–7558. Prerequisite pointer invoked at proof line(s) 499; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme and let X be an algebraic space over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $x \in |X|$. Let $d \in \{0, 1, 2, \ldots, \infty\}$. The following are equivalent*

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $length_{\mathcal{O}_{X,\overline{x}}} \mathcal{F}_{\overline{x}} = d$
(2) *for some étale morphism* $U \to X$ *with* U *a scheme and* $u \in U$ *mapping to* x *we have* $length_{\mathcal{O}_{U,u}} (\mathcal{F}|_U)_u = d$
(3) *for any étale morphism* $U \to X$ *with* U *a scheme and* $u \in U$ *mapping to* x *we have* $length_{\mathcal{O}_{U,u}} (\mathcal{F}|_U)_u = d$

Proof. Let $U \to X$ and $u \in U$ be as in (2) or (3). Then we know that $\mathcal{O}_{X,\overline{x}}$ is the strict henselization of $\mathcal{O}_{U,u}$ and that

$$\mathcal{F}_{\overline{x}} = (\mathcal{F}|_U)_u \otimes_{\mathcal{O}_{U,u}} \mathcal{O}_{X,\overline{x}}$$

See Properties of Spaces, Lemmas 04KF and 05VP. Thus the equality of the lengths follows from Algebra, Lemma 02M1 the fact that $\mathcal{O}_{U,u} \to \mathcal{O}_{X,\overline{x}}$ is flat and the fact that $\mathcal{O}_{X,\overline{x}}/\mathfrak{m}_u \mathcal{O}_{X,\overline{x}}$ is equal to the residue field of $\mathcal{O}_{X,\overline{x}}$. These facts about strict henselizations can be found in More on Algebra, Lemma 07QM. □